

Targeted interventions and prediction mistakes

Lecture 13 – AI4H

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Where are we at?

- Part I: Foundations
 - Tree-based, deep learning, transformers, LLMs
- Part II: Policy and ML
 - *Fairness*
 - Prediction policy
 - AI for policy, other uses
- Part III: Causal ML
 - Causal trees and forests
 - Double ML
 - Applications of causal forest etc

Takeaways of causal ML

- ML for causal estimation
 - New way of estimating heterogeneity
 - Causal forest etc
 - New way of isolating causal effect
 - Double machine learning

Overview of this lecture

- Targeted interventions through ML-based heterogeneity
 - Household energy savings
 - Active labor market policies
- Mistakes in prediction policy
- Research proposal questions and ideas

Targeted interventions with ML

Household energy savings

Knittel & Stolper (2025)

Motivation

- Behavioral energy nudges (Home Energy Reports, HERs) widely used (Opower).
- Large variation in responsiveness across households → opportunity for **ML-driven targeting**.
- Two goals:
 - Estimate heterogeneity using **causal forests**.
 - Evaluate **welfare gains** from targeted treatment.

OPower intervention

- Personalized reports of household consumption
 - Comparison to peers and plans
- Rollout
 - 15 RCT waves, ~900k households.
 - Different states,
 - Hard-copy vs email HERs; baseline-heavy targeting by the utility.
 - Controls well-balanced in each wave

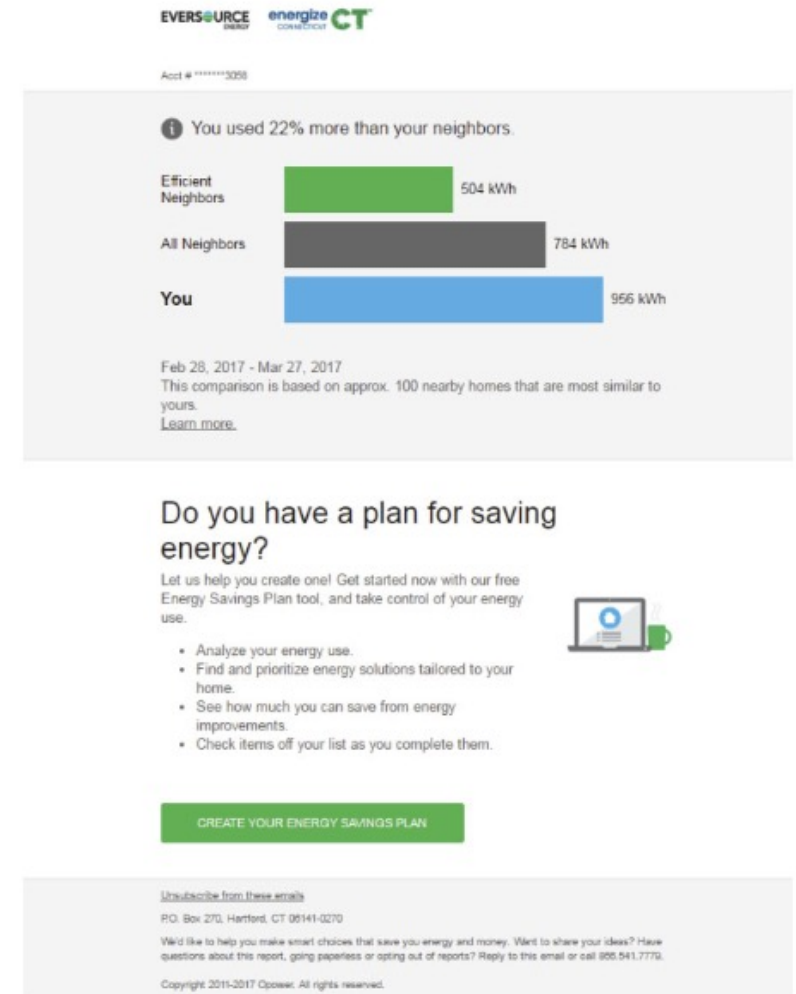


Fig. 1. *Eversource Home Energy Report.*
Source: Eversource.

Measuring the impact

- Average:

$$\text{kWh}_{it} = \alpha_1 + \alpha_2 T_{it} + X_i \eta + \theta_i + \omega_t + \varepsilon_{it}$$

- Conditional average:

$$\text{kWh}_{iwt} = \beta_1 + \sum_{j=-12}^{36} (\tau^j D_{iwt}^j) + X_i \eta + \theta_i + \omega_{wt} + \varepsilon_{iwt}$$

Average impact

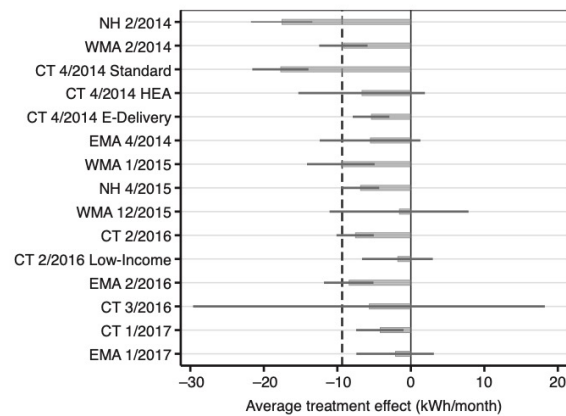


Fig. 3. *Average Treatment Effects, by Wave: Consumption.*

Notes: The y axis denotes a specific wave (waves are ordered by start date), and the x axis measures the ATE in kilowatt hours. The vertical dashed line denotes the weighted pooled average of all wave-specific ATEs, with weights equal to the wave sample size. Error bars denote 95% confidence intervals. CT, Connecticut; EMA, Eastern Massachusetts; NH, New Hampshire; WMA, Western Massachusetts. All effects are estimated via a panel difference-in-differences regression, using (1) and as described in Section 2.1.

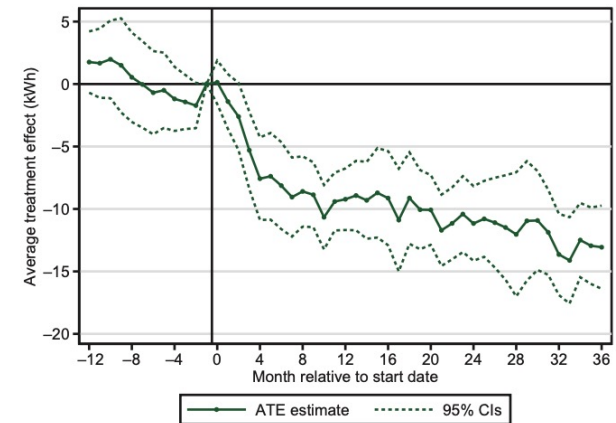


Fig. 4. *Event Study of Pooled Experimental Waves: Consumption.*

Notes: The solid line data points are event-study coefficients from estimation of (2). Dashed lines indicate 95% confidence intervals. The event-study binary variable corresponding to the month immediately prior to the start of HER mailings is omitted from the regression, and thus set to zero in the figure; all other points are interpretable as predictive effects relative to this omitted month.

CATE distribution

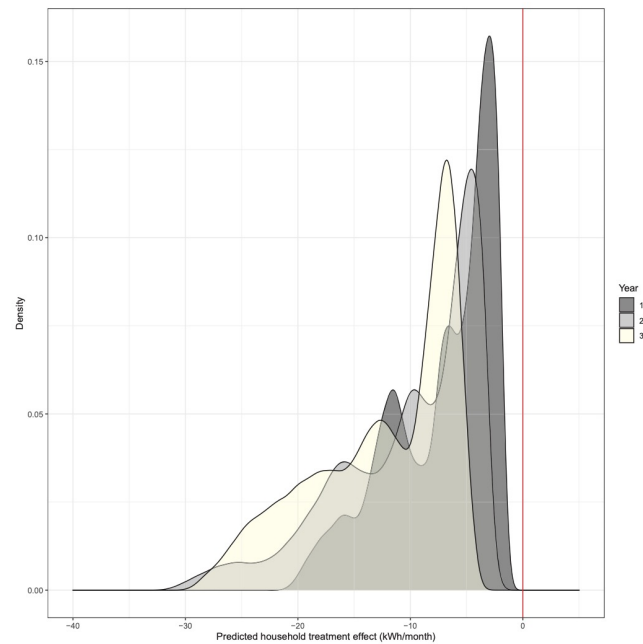


Fig. 5. *Distribution of Predicted Treatment Effects.*

Notes: Each plotted distribution is a kernel density of household treatment effects in a specific year (1, 2 or 3) of HER programming. The sample is fixed across years: only households with non-missing consumption in all three post-years are included. Treatment effect predictions come from our causal forest (Section 2.2).

Explaining CATE

Table 3. Characteristics of High versus Low Treatment Effect Households.

	Top 20% (1)	Bottom 20% (2)	Difference (3)
Social comparison (kWh)	353.02	-150.51	503.52***
Baseline consumption (kWh)	1,260.83	782.84	477.99***
Home value (\$)	355,758.96	460,184.34	-104,425.38***
Home square footage (100s)	20.62	21.04	-0.41
Number of rooms in home	7.26	7.28	-0.02
Year home built	1969.81	1967.78	2.03***
Single-family occupancy (= 1)	0.86	0.91	-0.05***
Renter (= 1)	0.1	0.1	-0.01
Annual income	102,908.33	107,962.35	-5,054.02*
Education (1-5)	3.22	3.35	-0.13***
GreenAware score (1-4)	2.21	2.17	0.03
Number of adults	2.75	2.43	0.32***
Child in home (= 1)	0.45	0.55	-0.10***
Participated in home energy audit	0.37	0.36	0.00
Age	58.07	54.79	3.27***

Notes: Columns (1) and (2) display mean characteristics among households in the top 20% and bottom 20%, respectively, of the predicted treatment effect distribution. 'Top 20%' indicates the largest reducers (of electricity consumption); 'Bottom 20%' indicates the smallest reducers (as well as any increasers). Predictions are generated through a procedure developed by Chernozhukov *et al.* (2023) and implemented by Deryugina *et al.* (2019); see [Online Appendix C.2](#) for details. Column (3) displays differences between columns (1) and (2), along with SEs clustered by zip code in parentheses, estimated via regression of each characteristic separately on a binary variable equalling one if a household is in the top quintile. * $p < .1$, *** $p < .01$.

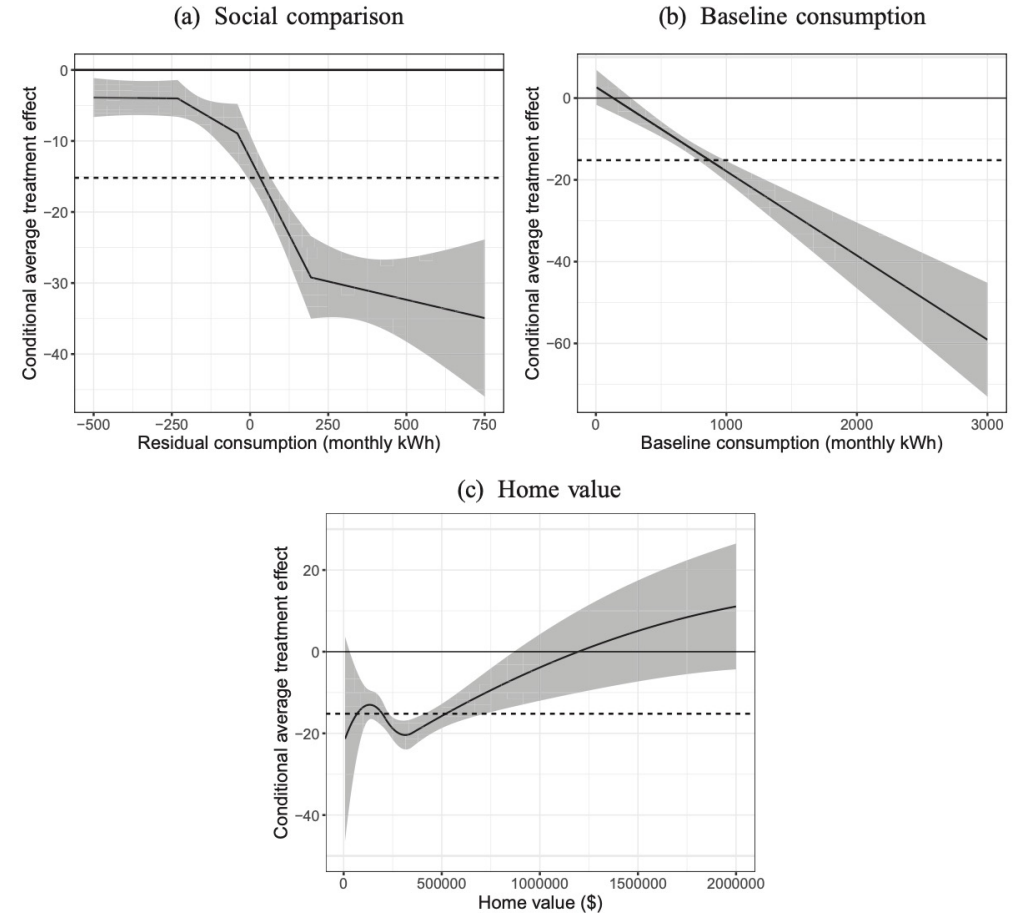


Fig. 7. Predicted Treatment Effect versus Household Type.

Notes: The x axis measures a different household attribute in each panel: (a) the social comparison variable, (b) baseline consumption and (c) home value. The y axis measures the predicted treatment effect as a function of the relevant attribute. The solid line and confidence intervals are estimated via spline regression, using the causalDML R package (Knaus, 2022). The dashed line is a point estimate of the average treatment effect. The sample includes all households with non-missing consumption in the year prior to the program start date and each of the first three years following the program start date. See [Online Appendix C.3](#) for further implementation details.

Targeted interventions

- Who to treat?
 - Cost of sending HER is 7\$
 - Treat all with net positive

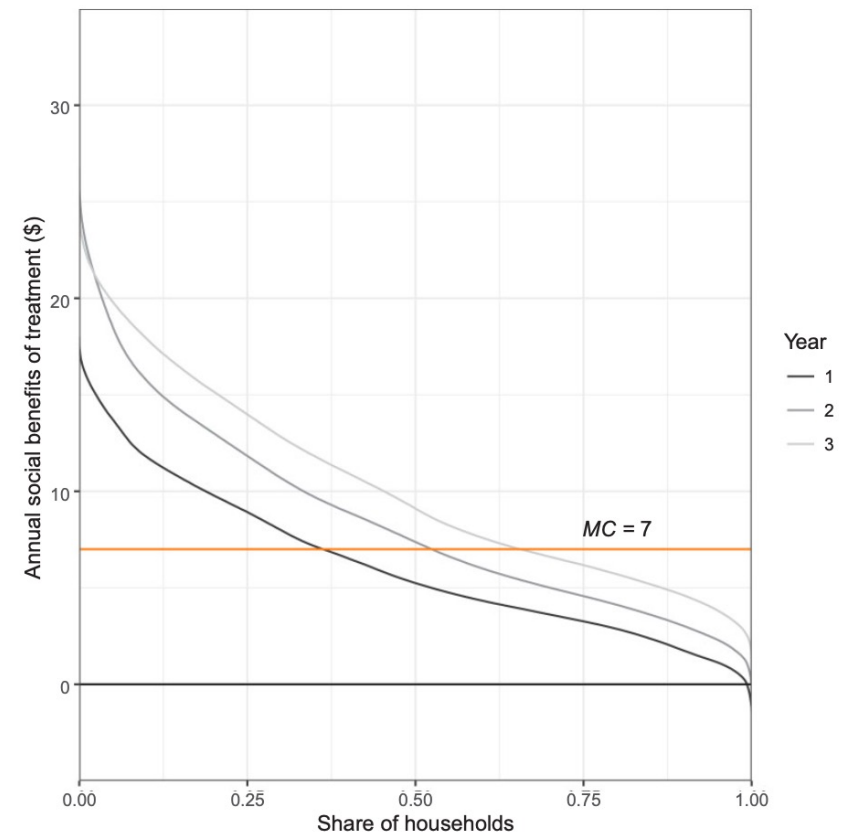


Fig. 8. *The Predicted Cumulative Distribution of HER-Induced Social Benefits.*

Notes: Each downward-sloping line is the reverse cumulative distribution function of annual social benefits produced by households in a given HER program year, estimated via our causal forest. The sample is fixed across years: only households with non-missing consumption in all three post-years are included. The line labelled ' $MC = 7$ ' denotes the estimated marginal cost of sending one year's worth of HERs to a household.

Active labor market policies

Goller et al. (2025)

Motivation & Research Questions

- Long-term unemployment
 - persistent issue in Germany (and other countries!)
- Job centers allocate ALMPs
 - lack evidence on **heterogeneous** effects
- Questions:
 - Do programs affect worker outcomes?
 - How heterogeneous are impacts (overall / by gender)?
 - Can CATE-based estimates improve **program allocation**?

Programs & Institutional Setup

- Overview of schemes for activation and integration:
 - Job-training (JT)
 - Program by external provider to help with crafting job applications
 - Reducing impediments (RIM)
 - Program by external provider to help with knowledge / motivation
 - Placement services (PS)
 - Direct contact to firms
 - In-firm training (IFT)
- Short program durations
 - JT/RIM: < 3 months (90%)
 - PS / IFT: < 1 month;

Data Overview

- Administrative universe of German UB II recipients (~300k individuals).
- Key descriptive differences
 - Participants have better recent employment histories.

Table 1
Descriptive statistics – outcome and selected covariates.

Variable	Men		Women	
	Participants	Non-participants	Participants	Non-participants
Cumulated days in regular employment in the 3 years after treatment (Outcome)	250 (325)	181 (291)	188 (300)	137 (261)
Personal characteristics				
Age at sampling date (in years)	38.3 (8.52)	39.7 (8.57)	38.9 (8.38)	39.9 (8.41)
Days since last employment	1478 (1598)	1822 (1774)	1949 (2128)	2115 (2287)
Days in regular employment in the previous 5 years	293 (401)	224 (358)	178 (343)	138 (298)
Days in welfare receipt in the last year	297 (112)	318 (91)	317 (98)	332 (79)
Receiving income from dependent employment (yes=1)	0.17	0.18	0.22	0.27
Region (west=0, east=1)	0.25	0.35	0.26	0.33
Foreigner (yes=1)	0.23	0.20	0.20	0.22
Job center characteristics				
Job center district - client-staff ratio	158 (28)	162 (26)	159 (28)	162 (27)
Sanction intensity due to violations of duties (in %)	0.54 (0.25)	0.51 (0.24)	0.54 (0.25)	0.52 (0.24)
Sanction intensity due to failure in reporting (in %)	0.71 (0.25)	0.71 (0.25)	0.71 (0.26)	0.71 (0.26)
District-level characteristics				
District unemployment rate (in %)	10.4 (3.3)	11.0 (3.5)	10.3 (3.3)	10.8 (3.6)
District unemployment rate of welfare recipients (in %)	7.4 (3.0)	7.9 (3.3)	7.4 (3.0)	7.7 (3.3)
N	28,425	136,691	20,741	116,769

Note. – Means of the covariates. Standard deviations are in parentheses. The treated group in this table contains all individuals from our initial three treatment groups (*JT*, *RIM*, and *PS*). We computed the mean of covariates over all treatment groups here.

Note on selection

- Caseworkers assign programs based on rich profiling data (much of which is observed).
 - Strict sanctions reduce program refusal.
 - $\Rightarrow$ Selection is non-random, but plausibly captured by observables
 - Plausibly identified
 - Supported using placebo test (except IFT)

Average treatment effects (ATE)

Table 2
Average treatment effects.

	Men					Women				
	NP	JT	RIM	PS	ATET	NP	JT	RIM	PS	ATET
Cumulated Days in regular Employment in 36 months after start of treatment (outcome)										
NP	181.1 (2.1)					137.6 (2.2)				
JT	36.3*** (5.4)	217.4 (4.9)			35.0*** (6.2)	25.7*** (5.6)	163.3 (5.1)			23.6*** (5.9)
RIM	34.6*** (6.3)	-1.7 (7.8)	215.7 (6.2)		34.1*** (6.8)	36.8*** (6.7)	11.2 (8.3)	174.4 (6.4)		34.7*** (6.9)
PS	45.1*** (7.8)	8.8 (9.0)	10.5 (9.6)	226.2 (7.0)	46.7*** (7.6)	54.1*** (8.7)	28.5*** (10.0)	17.3 (10.7)	191.7 (8.4)	64.1*** (9.1)
N	136,691	12,329	8721	7375		116,769	9350	6678	4713	

Note. – Outcomes are measured in days in regular employment after starting the treatment. ATET relative to no treatment only; Potential outcomes are on the main diagonals. Standard errors are in parentheses. *** indicates that the p-value of a two-sided significance test is below 1 %. The programs are labeled as NP: non-participation, JT: job-training, RIM: reducing impediments, and PS: placement services.

- Takeaway:
 - all three programs are effective, but
 - potential for match effects?

Group average treatment effects (GATE)

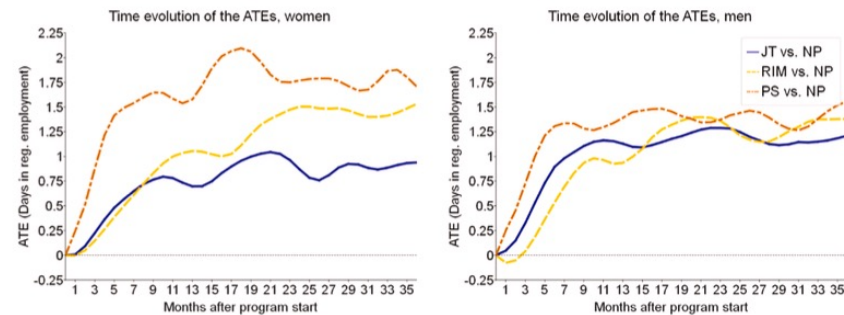


Fig. 1. Evolution of ATE over time. Note. – Outcome is the cumulative days in regular employment for each month after the start of participation in the respective program. After month 4, all estimates significantly differ from zero at conventional levels. Exemplary standard errors for month 10 (women: 0.19, 0.23, 0.31; men: 0.19, 0.23, 0.28), month 20 (women: 0.24, 0.28, 0.37; men: 0.23, 0.27, 0.33), and month 30 (women: 0.25, 0.30, 0.39; men: 0.24, 0.28, 0.33) for JT vs. NP, RIM vs. NP, and PS vs. NP, respectively.

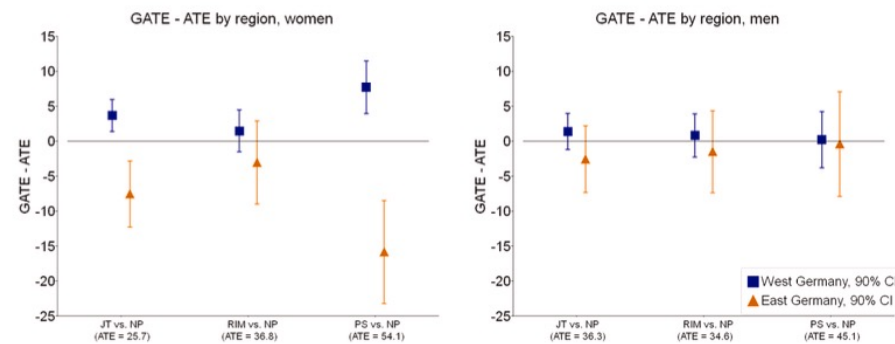


Fig. 2. Difference of GATEs to ATE in West / East Germany. Note. – 90 % confidence intervals (CI). ATE estimates are in parentheses at the bottom of the graphs. The outcome is the cumulative days in regular employment in the 36 months after the start of treatment.

Modified Causal Forest

- They use the Modified Causal Forest
 - Stable with multiple treatments and many groups).
 - Recall - 3 programs + non-assignment
- Maintains honest estimation + treatment-oriented splitting rules.

Individual average effects (IATE)

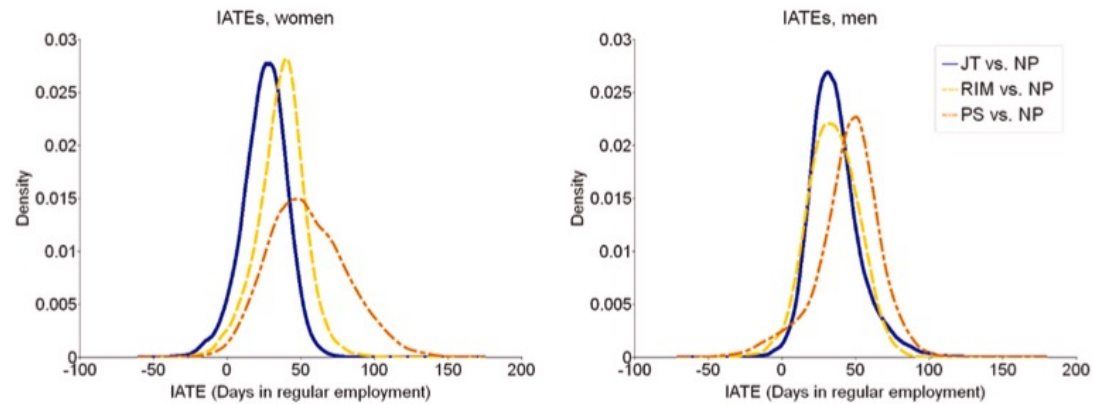


Fig. 6. Distribution of estimated IATEs. Note. – IATE density plots. The outcome is cumulative days in regular employment in 36 months after the start of treatment.

Targeted interventions

Table 6
Overall effects of simulated hypothetical program allocations.

Overall effects of simulated hypothetical program allocations					
	Share in different programs (in %)			Cum. # of days in reg. empl. in 36 months after start of treatment	Gain for switchers (in %)
	JT	RIM	PS		
Panel A: Hypothetical program allocations					
Observed	7.20	5.09	3.95	171.25	-
Random	7.26	5.06	3.84	171.26	+ 0.00
Policy simulation					
- No constraint	10.21	22.79	66.91	222.39	+ 31.27
- No constraint, only significant	10.13	14.96	51.03	208.58	+ 35.54
- Constrained, preference to largest gain	7.20	5.09	3.95	177.35	+ 12.51
- Constrained, sequential optimization	7.20	5.09	3.95	178.16	+ 14.22
- Constrained, preference to days since last empl.	7.20	5.09	3.95	172.03	+ 1.54
Panel B: Decision Trees					
- Constrained, 2 level	6.58	4.40	3.82	172.48	+ 2.40
- Constrained, 3 level	7.35	4.87	3.85	174.03	+ 5.42
- Constrained, 4 level	7.33	4.93	3.72	174.40	+ 6.12

Note. – Share in non-participation can be calculated as 100 % - Share JT / RIM / PS. In the “Observed” allocation, the outcome is the realized mean of cumulated days in regular employment 36 months after the start of treatment. In all the other scenarios, some individuals are hypothetically switched from the actual to another treatment status. The “gain for switchers” reflects the gains in percent for those switching the treatment status; thus, it can be higher for the same absolute gain if fewer people are reallocated. The “Constrained” allocations take the shares as in the “Observed” allocation as given to simulate a budget constraint. In Panel B, this can deviate since constraints are binding only in the training set. n level refers to the number of levels in the decision tree, i.e., level 4 (or depth 4) can have at most 16 “leaves.”

Prediction mistakes

Identification with observation data – Rambachan (2024)

Motivation

- Observational data confounds three factors
 - preferences, beliefs, and private information.
- Selective labels problem
 - Addresses private info
 - We only observe outcome when $C=1$ (e.g., see detainee behavior only if released).
- Existing empirical work compares humans to algorithms but lacks an econometric framework to distinguish the factors
 - Goal of the paper: Identify and test for systematic prediction mistakes under weak assumptions.

Conceptual innovation

- Partition characteristics into included (x_I) and excluded (x_E)
 - x_I can directly affect utility
 - x_E can only affect beliefs - utility exclusion restriction
- Testable implications: Compute all prediction rules from data to create bounds on potential prediction mistakes.
 - In other words, systematic prediction mistakes: choices cannot be rationalized by any combination of utility, accurate beliefs and private information

Revealed prediction mistakes

- Under linear utility and accurate beliefs, judge/DM must use an (incomplete) threshold rule in expected outcome.
- Observable condition (Theorem 1):
 - For fixed x_I , we compare two utilities
 - the highest among released types (μ_1)
 - lowest upper-bound among detained types ($\bar{\mu}_0$) under *any* missing information
 - A misranking violates this: some released group has *higher* risk than worst-case detained group.
- Misranking $\Rightarrow$ systematic prediction mistakes.

When are mistakes identifiable

- If **all** characteristics affect utility ($x_I = x$), choices are always rationalizable
 - Then *mistakes untestable*.
- If missing-data bounds are uninformative (the unobserved distribution of outcome can be anything), also untestable.
- Thus, identification requires both:
 - Exclusion restriction on utility,
 - Informative bounds on unobserved outcomes.
- Empirical takeaway: Prediction-mistake detection is only possible under interpretable restrictions.

How to get informative bounds

- Use quasi-random judge assignment as an instrument.
- Conditional independence of outcome and instrument under covariates
 - Judges can have different utility, different private info
- Leverage instruments
 - produce sharp bounds on the identified set for $\bar{\mu}_0(x, z)$ (*Proposition 1*).
 - feed directly into the misranking test (*Proposition 2*).

Takeaways from today

- Use of causal ML for targeted interventions
 - Energy policy
 - Uncover large heterogeneities, examples:
 - Role of social comparison, house size
 - Targeted policy welfare improving
 - Improvement of assignment to active labor market program
 - 14 % under sequential assignment
 - 6% under interpretable trees
- Rambachan's framework – useful for measuring prediction mistakes

Course evaluation

Please take a couple of minutes to evaluate course

THANK YOU!

Your ideas / questions

- Are mathematical equations necessary?
- ??